main.c

Run ▶

```c
#include <stdio.h>
#include <string.h>

struct Symbol {
    char name[30];
    char type[20];
    char value[30];
};

struct Symbol table[100];
int count = 0;

void insert() {
    char name[30];

    printf("Enter symbol name: ");
    scanf("%29s", name);

    for (int i = 0; i < count; i++) {
        if (strcmp(table[i].name, name) == 0) {
            printf("Symbol already exists!\n");
            return;
        }
    }

```

Output

```
--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 1
Enter symbol name: x
Enter data type: int
Enter value: 10
Symbol inserted successfully!

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 2

Symbol Table:
Name            Data Type       Value
x               int             10
```

main.c

Run ▶

```
25
26      strcpy(table[count].name, name);
27
28      printf("Enter data type: ");
29      scanf("%19s", table[count].type);
30
31      printf("Enter value: ");
32      scanf("%29s", table[count].value);
33
34      count++;
35      printf("Symbol inserted successfully!\n");
36  }
37
38  void display() {
39      if (count == 0) {
40          printf("Symbol table is empty!\n");
41          return;
42      }
43
44      printf("\nSymbol Table:\n");
45      printf("%-15s %-15s %-15s\n", "Name", "Data Type", "Value");
46
47      for (int i = 0; i < count; i++) {
48          printf("%-15s %-15s %-15s\n",
49                 table[i].name,
```

Output

```
--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 3
Enter symbol to search: x
Symbol found!
Name: x
Data Type: int
Value: 10

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 4
Enter symbol to delete: x
Symbol deleted successfully!

--- SYMBOL TABLE OPERATIONS ---
1. Insert
```

```
                table[i].type,
                table[i].value);
    }
}

void search() {
    char name[30];

    printf("Enter symbol to search: ");
    scanf("%29s", name);

    for (int i = 0; i < count; i++) {
        if (strcmp(table[i].name, name) == 0) {
            printf("Symbol found!\n");
            printf("Name: %s\n", table[i].name);
            printf("Data Type: %s\n", table[i].type);
            printf("Value: %s\n", table[i].value);
            return;
        }
    }

    printf("Symbol not found!\n");
}

void deleteSymbol() {
    char name[30];
```

```
Enter symbol to search: x
Symbol found!
Name: x
Data Type: int
Value: 10

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 4
Enter symbol to delete: x
Symbol deleted successfully!

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 5
Exiting program.
```

main.c

Run ▶

```
    char name[30];

    printf("Enter symbol to delete: ");
    scanf("%29s", name);

    for (int i = 0; i < count; i++) {
        if (strcmp(table[i].name, name) == 0) {
            for (int j = i; j < count - 1; j++) {
                table[j] = table[j + 1];
            }

            count--;
            printf("Symbol deleted successfully!\n");
            return;
        }
    }

    printf("Symbol not found!\n");
}

int main() {
    int choice;

    do {
        printf("\n--- SYMBOL TABLE OPERATIONS ---\n");
        printf("1. Insert\n");
```

Output

```
Enter symbol to search: x
Symbol found!
Name: x
Data Type: int
Value: 10

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 4
Enter symbol to delete: x
Symbol deleted successfully!

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 5
Exiting program.
```

main.c

main.c

```c
        printf("2. Display\n");
        printf("3. Search\n");
        printf("4. Delete\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                insert();
                break;
            case 2:
                display();
```

main.c

Run ▶

```
        printf("2. Display\n");
        printf("3. Search\n");
        printf("4. Delete\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                insert();
                break;
            case 2:
                display();
                break;
            case 3:
                search();
                break;
            case 4:
                deleteSymbol();
                break;
            case 5:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice!\n");
```

Output

```
Enter symbol to search: x
Symbol found!
Name: x
Data Type: int
Value: 10

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 4
Enter symbol to delete: x
Symbol deleted successfully!

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 5
Exiting program.
```

main.c

Run ▶

```
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                insert();
                break;
            case 2:
                display();
                break;
            case 3:
                search();
                break;
            case 4:
                deleteSymbol();
                break;
            case 5:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice!\n");
        }
    } while (choice != 5);

    return 0;
}
```

Output

```
Enter symbol to search: x
Symbol found!
Name: x
Data Type: int
Value: 10

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 4
Enter symbol to delete: x
Symbol deleted successfully!

--- SYMBOL TABLE OPERATIONS ---
1. Insert
2. Display
3. Search
4. Delete
5. Exit
Enter your choice: 5
Exiting program.
```